

Clinical Risk Assessment Report: Patient 24

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Positive for Metastasis
- Prediction Confidence: High Confidence
- Absolute Predicted Risk: 43.6% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 1.97x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as High-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Large Zone Emphasis (CT) (GLZLM_LGZE.CT), which elevated the patient's absolute risk by +8.7%. Biologically, this feature reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

2. Key Pathological & Imaging Drivers (Elevating Risk)

Large Zone Emphasis (CT) (GLZLM_LGZE.CT)	Value: -0.42 [Bottom 1%] (+8.7%)
<i>Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.</i>	
Short Run High Gray-Level Emphasis (CT) (GLRLM_SRHGE.CT)	Value: 1.32 [Top 3%] (+4.0%)
<i>Reflects fragmented, short streaks of high density or metabolism, indicating a disjointed and aggressive tumor architecture.</i>	
Large Zone Low Gray-Level Emphasis (CT) (GLZLM_LZLGE.CT)	Value: -0.02 [Top 13%] (+2.9%)
<i>Indicates the dominance of large regions with low attenuation or metabolism, pointing to extensive necrosis or ground-glass opacities.</i>	
75th Percentile Tumor Density (CONVENTIONAL_HUQ3)	Value: 0.84 [Top 19%] (+2.5%)
<i>Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.</i>	
High Gray-Level Zone Emphasis (CT) (GLZLM_HGZE.CT)	Value: 1.53 [Top 2%] (+2.1%)
<i>Highlights the distribution of high-density or highly metabolic active regions clustered within the tumor.</i>	
Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT)	Value: -0.26 [Bottom 3%] (+1.9%)
<i>Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.</i>	
Squamous Cell Carcinoma Antigen (SCC_Ag)	Value: 0.7 [Avg (28th %ile)] (+1.2%)
<i>Serum levels reflect tumor proliferation, burden, and squamous cell differentiation.</i>	
Patient Age (Age)	Value: 53.0 [Bottom 16%] (+1.0%)
<i>Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.</i>	

3. Protective / Mitigating Factors (Reducing Risk)

No major risk-reducing features observed in the top drivers.

Machine Learning Feature Attribution (SHAP Decision Path)

